

Linear Regression of an Indicator Matrix

- Suppose there are K classes and for each individual we have an indicator of its class $\underline{y}_i = (y_{i1}, \dots, y_{iK})$ where

$$y_{ih} = \begin{cases} 1 & \text{if } i \text{ belongs to class } h. \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

- Combining that information for each individual, we get the indicator response matrix

$$Y = \begin{pmatrix} \underline{y}_1 \\ \vdots \\ \underline{y}_N \end{pmatrix} \in \mathbb{R}^{N \times K}. \quad (2)$$

- We need a coefficient matrix $B \in \mathbb{R}^{p \times K}$ minimizing RSS across all response columns $h = 1, \dots, K$. The loss function is:

$$\text{RSS}(B) = \sum_{h=1}^K \left(\underline{y}_h - X\beta_h \right)' \left(\underline{y}_h - X\beta_h \right) \quad (3)$$

where $\underline{y}_h$ is the h -th column of Y s.t. $\underline{y}_h \in \mathbb{R}^N$, X is the design matrix $\in \mathbb{R}^{N \times p}$ and $\beta_h \in \mathbb{R}^p$ is the h -th column of B . Note that X includes a column of 1's for the intercept.

- Since the loss function is a sum of K independent terms, we can minimize each term independently, leading to:

$$\hat{\beta}_h = (X'X)^{-1}X'\underline{y}_h, \quad \text{so that:} \quad (4)$$

$$\hat{B} = \left[(X'X)^{-1}X'\underline{y}_1 \mid \dots \mid (X'X)^{-1}X'\underline{y}_K \right] = (X'X)^{-1}X' \left[\underline{y}_1 \mid \dots \mid \underline{y}_K \right] \quad (5)$$

$$\Rightarrow \hat{B} = (X'X)^{-1}X'Y, \quad \text{and the fitted values are } \hat{Y} = X(X'X)^{-1}X'Y. \quad (6)$$

- A new observation with input $x \in \mathbb{R}^p$ (whose first term is a 1) is classified as follows:
 - (i) Compute the fitted value $\hat{f}(x) = x' \hat{B} \in \mathbb{R}^{1 \times K}$ (row vector).
 - (ii) Identify the largest component and classify accordingly:

$$\hat{G}(x) = \arg \max_{h=1, \dots, K} \hat{f}_h(x). \quad (7)$$

- The rationale for this approach is to view the regression as an estimate of conditional expectation.
- For Y_h , we get: $\mathbb{E}(Y_h \mid X = x) = \mathbb{P}(Y_h \mid X = x)$.
- However, how good of an approximation is the linear regression model?
- A nice property of this model is that since $\sum_{h=1}^K Y_h = 1 \quad \forall i = 1, \dots, N$ we have:

$$\begin{pmatrix} \sum_{h=1}^K y_{1h} \\ \vdots \\ \sum_{h=1}^K y_{Nh} \end{pmatrix} = \sum_{h=1}^K \underline{y}_h = \underline{1}_N, \quad \text{so that:} \quad (8)$$

$$\sum_{h=1}^K \hat{\underline{y}}_h = \sum_{h=1}^K X(X'X)^{-1} X' \underline{y}_h = X(X'X)^{-1} X' \sum_{h=1}^K \underline{y}_h = X(X'X)^{-1} X' \underline{1}_N \quad (9)$$

$$= P_X \underline{1}_N, \quad \text{where } P_X \text{ is the orthogonal matrix projection matrix onto the space generated by the features.} \quad (10)$$

$$\left(\equiv \hat{f}_h(x) \text{ above} \right) \quad (11)$$

- Because $\underline{1}_N$ is the first column of X and $P_X X = X$ (the vector $\underline{1}_N$ is already in the column space of X), we have $P_X \underline{1}_N = \underline{1}_N$ and hence $\sum_{h=1}^K \hat{y}_h = \underline{1}_N$.
- In that sense, $\hat{f}_h$ meets the requirement of summing to 1 as long as there is an intercept in the model. Yet there is no requirement on the sign and magnitude, so that it can be negative or greater than 1.
- A problem occurs when $K \geq 3$, consider the following example:

$$Y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = (\underline{y}_1 \quad \underline{y}_2 \quad \underline{y}_3) \in \mathbb{R}^{N=3 \times K=3} \quad (12)$$

$$X = \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \in \mathbb{R}^{N=3 \times p=2}, \quad B = \begin{pmatrix} \beta_{11} & \beta_{12} & \beta_{13} \\ \beta_{21} & \beta_{22} & \beta_{23} \end{pmatrix} \quad (13)$$

- Then:

$$\hat{B} = (X'X)^{-1} X'Y = \left[\begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \right]^{-1} \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix} \quad (14)$$

$$= \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix} \quad (15)$$

$$\Rightarrow \hat{B} = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ -1/2 & 0 & 1/2 \end{pmatrix} \quad (16)$$

- For a new input $x = \begin{pmatrix} 1 \\ x \end{pmatrix}$, the fitted value is:

$$\hat{f}(x) = (1 \quad x) \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ -1/2 & 0 & 1/2 \end{pmatrix} \quad (17)$$

$$= (1/3 - \frac{1}{2}x, \quad 1/3, \quad 1/3 + \frac{1}{2}x), \quad \text{this is our classifier.} \quad (18)$$

- To classify Class 2, we would need:

$$\begin{cases} 1/3 > 1/3 - \frac{1}{2}x \\ 1/3 > \frac{1}{2}x + 1/3 \end{cases} \iff \begin{cases} -\frac{1}{2}x < 0 \\ \frac{1}{2}x < 0 \end{cases} \iff x < 0 \quad \text{and} \quad x > 0, \quad \text{a contradiction.} \quad (19)$$

Hence Class 2 is completely masked by Class 1 and 3.